B) 9 days

C) 10 days

D) 12 days

E) 15 days

PIPES AND CISTERN

Question 1: Two pipes, A and B, can fill a tank separately in 12 and 16 hours, respectively. If both of them are opened together when the tank is initially empty, how much time will it take to completely fill the tank?

Question 2: Three pipes, A, B, and C, are connected to a tank. Out of the three, A and B are the inlet pipes, and C is the outlet pipe. If opened separately, A fills the tank in 10 hours, B fills the tank in 12 hours, and C empties the tank in 30 hours. If all three are opened simultaneously, how much time does it take to fill / empty the tank?

Question 3: Three pipes A, B, and C are connected to a tank. Out of the three, A is the inlet pipe and B and C are the outlet pipes. If opened separately, A fills the tank in 10 hours, B empties the tank in 12 hours and C empties the tank in 30 hours. If all three are opened simultaneously, how much time does it take to fill / empty the tank ?

Question 4: A cistern has two pipes. Both working together can fill the cistern in 12 minutes. First pipe is 10 minutes faster than the second pipe. How much time would it take to fill the cistern if only second pipe is used ?

Question 5: Three pipes A, B and C are connected to a tank. Out of the three, A and B are the inlet pipes and C is the outlet pipe. If opened separately, A fills the tank in 10 hours and B fills the tank in 30 hours. If all three are opened simultaneously, it takes 30 minutes extra than if only A and B are opened. How much time does it take to empty the tank if only C is opened?

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Question 6: Time required by two pipes A and B working separately to fill a tank is 36 seconds and 45 seconds respectively. Another pipe C can empty the tank in 30 seconds. Initially, A and B are opened and after 7 seconds, C is also opened. In how much more time the tank would be completely filled?

Question 7: Two pipes A and B can fill a tank in 20 hours and 30 hours respectively. If both the pipes are opened simultaneously, find after how much time should pipe B be closed so that the tank is full in 18 hours?

Question 8: 9 pumps working 8 hours a day can empty a reservoir in 20 days. How many such pumps needed to empty the same reservoir working 6 hours a day in 16 days.

Question 9: A leak can empty a completely filled tank in 10 hours. If a tap is opened in completely filled tank which admits 4 liters of water per minute, then leak takes 15 hours to empty the tank. How many litres of water does the tank hold

Question 10: A, B and C can fill a tank in 6 hours together. After working for 2 hours, C is closed and A and B fill it in 7 hours more. Find the time taken by C alone to fill the tank?

Question 11: A tap drives at a rate of one drop/second and 800 drops = 100 ml. The number of litres water wasted in 30 days of a month is ?

Question 12: Two pipes A and B independently can fill a tank in 20 hours and 25 hours. Both are opened together for 5 hours after which the second pipe is turned off. What is the time taken by first pipe alone to fill the remaining portion of the tank?

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Practice Questions

1. A tank can be filled by pipe A in 5 hours and by pipe B in 6 hours. If both pipes are opened together, how

long will it take to fill the tank?

2. A tank has two inlet pipes and one outlet pipe. The first inlet pipe can fill the tank in 4 hours, while the

second can fill it in 6 hours. The outlet pipe can empty the full tank in 12 hours. If all pipes are opened

simultaneously, how long will it take to fill the tank?

3. Pipe A can fill a tank in 10 hours, while pipe B can empty it in 15 hours. If both pipes are opened

simultaneously, how long will it take to fill the tank?

4. A cistern has three pipes. The first two can fill it in 10 hours and 12 hours respectively, while the third can

empty it in 15 hours. If all pipes are opened together, how long will it take to fill the cistern?

5. A tank can be filled by pipe A in 6 hours and emptied by pipe B in 8 hours. If both pipes are opened when

the tank is empty, how long will it take to fill the tank?

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6. Pipe A can fill a tank three times as fast as pipe B. If they can fill the tank together in 36 minutes, how long

will it take pipe B alone to fill the tank?

7. Two pipes can fill a tank in 15 hours and 12 hours respectively. Both pipes are opened for 4 hours, then the

first pipe is closed. How long will it take for the second pipe to complete filling the tank?

8. A tank has an inlet pipe and an outlet pipe. The inlet pipe can fill the tank in 4 hours, and the outlet pipe can

empty it in 6 hours. If both pipes are opened when the tank is half full, how long will it take to empty the tank

completely?

9. Three pipes A, B, and C can fill a tank in 6 hours, 8 hours, and 12 hours respectively. If all pipes are opened

together, how long will it take to fill the tank?

10. A cistern has two inlet pipes and one outlet pipe. The inlet pipes can fill the cistern in 8 hours and 12 hours

respectively, while the outlet pipe can empty it in 6 hours. If all pipes are opened when the cistern is empty,

how long will it take to fill 3/4 of the cistern?

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Practice Questions

1. A tank can be filled by pipe A in 5 hours and by pipe B in 6 hours. If both pipes are opened together, how long will it take to fill the tank?

2. A tank has two inlet pipes and one outlet pipe. The first inlet pipe can fill the tank in 4 hours, while the second can fill it in 6 hours. The outlet pipe can empty the full tank in 12 hours. If all pipes are opened simultaneously, how long will it take to fill the tank?

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5. A tank can be filled by pipe A in 6 hours and emptied by pipe B in 8 hours. If both pipes are opened when the tank is empty, how long will it take to fill the tank?

6. Pipe A can fill a tank three times as fast as pipe B. If they can fill the tank together in 36 minutes, how long will it take pipe B alone to fill the tank?

7. Two pipes can fill a tank in 15 hours and 12 hours respectively. Both pipes are opened for 4 hours, then the first pipe is closed. How long will it take for the second pipe to complete filling the tank?

8. A tank has an inlet pipe and an outlet pipe. The inlet pipe can fill the tank in 4 hours, and the outlet pipe can empty it in 6 hours. If both pipes are opened when the tank is half full, how long will it take to empty the tank completely?

9. Three pipes A, B, and C can fill a tank in 6 hours, 8 hours, and 12 hours respectively. If all pipes are opened together, how long will it take to fill the tank?

10. A cistern has two inlet pipes and one outlet pipe. The inlet pipes can fill the cistern in 8 hours and 12 hours respectively, while the outlet pipe can empty it in 6 hours. If all pipes are opened when the cistern is empty, how long will it take to fill 3/4 of the cistern?

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